

Homework 8

ORF 245 - Fundamentals of Statistics

Due: Monday Apr. 21st, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Commuting Times)

The U.S. Census Bureau determines how far commuters have to travel to work using a nation-wide survey. Based on an extremely large sample, the national average commute distance is 13 miles.

A real estate office wants to make a similar survey in a certain town. A simple random sample of 400 households is chosen, the occupants are interviewed, and it is determined that on the average, the commuters in sampled households traveled on average 13.7 miles to work, with a standard deviation of 12.0 miles.

- (a) Construct a lower one-sided 95% confidence interval for the mean commute time for the head of households in this town. Justify any approximations or assumptions you may have used in your computation.
 - (b) On average, do the heads of households in the town have to commute farther than their national counterparts? Perform an appropriate statistical test at significance level 5%. Your conclusion about whether to reject H_0 , or not should be clearly stated and motivated.
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Exercise: 2. (Type II Error and Minimum Sample Size)

For a type of paint, the mean drying time is 75 minutes. Princeton researchers have proposed a new additive designed to **decrease** the drying time.

An i.i.d. random sample of size $n = 80$ has sample mean $\bar{x} = 69.8$ minutes. Assume that the standard deviation of this population is **known** to equal 10 minutes. Let μ be the population average drying time of the paint with the new additive in minutes.

Let the significance level be $\alpha = 5\%$.

- (a) Is the new additive effective? Be sure to: i) formulate the null and alternative hypotheses, ii) propose a test statistic and rejection region for a test of size 5%, iii) compute the value of the test statistic for the given data, iv) compute the associated p-Value, and v) clearly report the outcome of the test.

- (b) What is the probability of a Type II Error of the test if the true value of average drying time with the additive is 70 mins? (i.e. if $\mu = 70$)

Hint: what is the distribution of the test statistic $T(\dots)$ from part (a) if the population mean is equal to 70? It should not be standard normal...

- (c) What is the minimum sample size required so that if $\mu = 70$, the probability of a Type II Error is at most 1%? Assume that we maintain the significance level α at 5%.

Exercise: 3. (Small-sample Hypothesis Tests)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, with $n < 30$.

We are given null hypothesis of $H_0 : \mu = \mu_0$, for some given value of μ_0 , and alternative hypothesis $H_A : \mu > \mu_0$. Let $\alpha \in (0, 1)$ denote the desired significance level for our test.

- (a) Assuming the variance σ^2 is **known**, find both: i) a test statistic $T(X_1, \dots, X_n)$ and ii) a rejection region R for a test of size α . Justify your answer.

Be sure to clearly state your test statistic and its distribution under the null hypothesis.

- (b) Now, assume the variance σ^2 is **unknown**. Find a test statistic and rejection region for a test of size α . Again, justify your answer.

As before, be sure to clearly state your test statistic and its distribution under the null hypothesis.

Exercise: 4. (Simple Linear Regression)

For this question, you may compute by calculator or using R, but you may not use the function `lm`. Consider the data in the below table (also found in companion file `data.csv`).

- (a) Compute the following sample statistics:

- $\bar{x}, \bar{y}$ sample means of x, y
- $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$
- $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

- (b) Assume that the data comes from the model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, and all our usual assumptions are satisfied (i.i.d data, exogeneity, normally distributed errors).

Calculate the least squares (i.e. ‘best fit’) estimate for β_0, β_1 , denoted $\hat{\beta}_0, \hat{\beta}_1$.

- (c) Compute the sum of squared errors (SSE) and the residual standard error $\hat{\sigma}_\epsilon$

x	y
99.0	28.8
101.1	27.9
102.7	27.0
103.0	25.2
105.4	22.8
107.0	21.5
108.7	20.9
110.8	19.6
112.1	17.1
112.4	18.9
113.6	16.0
113.8	16.7
115.1	13.0
115.4	13.6
120.0	10.8

- (d) Calculate the standard error of $\hat{\beta}_1$, denoted $SE(\hat{\beta}_1)$. Can we reject the null hypothesis $H_0 : \beta_1 = 0$ in favor of the alternative hypothesis $H_A : \beta_1 \neq 0$ at 5% significance level? You may assume that $\hat{\beta}_1$ is normally distributed around β_1 with variance $SE(\hat{\beta}_1)^2$

Exercise: 5. [OPTIONAL] (Two-sample Hypothesis Test)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

The level of lead in the blood (mcg/dL) was determined for a sample of 152 NJ male hazardous-waste workers and also for a sample of 86 NJ female workers.

The sample mean is 5.5 and 3.8 for male and female respectively. The sample variance for male and female is also found to be 13.68 and 3.44.

Let $\mu_d = \mu_{male} - \mu_{female}$ be the difference between true average blood lead levels for male and female workers.

- Is there substantial difference in the mean level of lead in the blood between these two groups? Formulate the null hypothesis and alternative hypothesis, compute the p-value and answer the question using $\alpha = 0.05$.
- What is the power of the test if the difference μ_d is $\Delta' = 0.5$? What does this mean?
- Construct a two-sided 95% confidence interval for the difference between true average lead levels of blood for male and female workers.
- What assumptions do you make in the above calculations?